

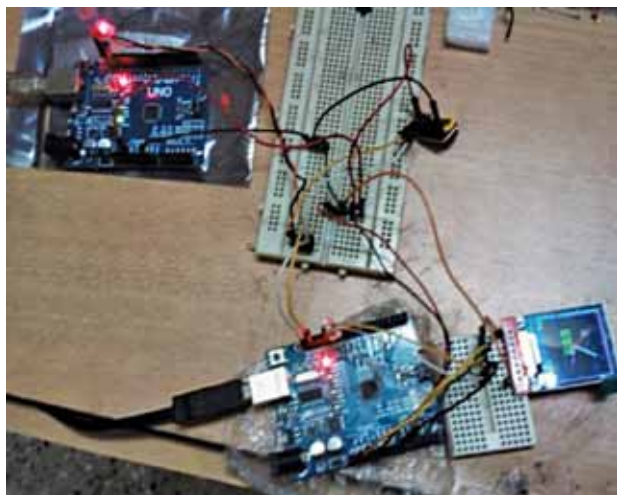


Fig. 2: Prototype-1



Fig. 3: Prototype-2

`unsigned long now_time = hour*10000 + minute*100 + second;`
for example, 72323 is 7:23:23

TFT ILI9163 is an 128x128 dots colour display that runs on SPI bus along with Reset, AO, and CS

PARTS LIST

Semiconductors:

Board1, Board2 - Arduino Uno board
T1, T2 - BC547 npn transistor
D1, D2 - 1N4007 rectifier diode
- DS3231 RTC module

Resistors (all 1/4-watt, ±5% carbon):

R1, R2 - 3.3-kilo-ohm
R3 - 470-ohm

Miscellaneous:

CON1 - 2-pin connector
CON2 - 6-pin connector
CON3 - 8-pin connector
CON4, CON5 - 3-pin connector
RL1, RL2 - 5V, 1CO sugar-cube relays
TFT1 - 128x128 ILI9163 TFT
- Jumper wires
- 5V regulated power supply

pins that run on the second Arduino. When the GPIO pin fires on the first Arduino (Board1), the same is read by the second Arduino (Board2) on pin A3 using `digitalRead` (relay) to trigger the relay on/relay off display.

Schematic

Digital pins 3 and 4 act as relay control pins from the first Arduino while the A3 pin on the second Arduino reads this status to process inside the second Arduino. TFT ILI9163 is connected on the second Arduino to display time. The RTC DS3231 is shared by both these Arduinos. The relays (RL1 and RL2) connection go further to the contactors for switching off the AC, which needs to handle bigger load.

Depending on the triggering action in the program, both relays RL1 and RL2 will be energised or deenergised simultaneously.

Following relay on/off time formats are used in the source code:

```
//----- Enter on times -----
int on_hour[] = {12,12,12,12,12,12,12,12,12,12,12,12};
int on_min[] = {15,17,19,22,23,25,27,29,31};
int on_sec[] = {15,15,15,15,15,15,15,15,15};
//----- Enter off times -----
```

```
int off_hour[] = {12,12,12,12,12,12,12,12,12,12,12,12};
int off_min[] = {16,18,20,22,24,26,28,30,32};
int off_sec[] = {15,15,15,15,15,15,15,15,15};
```

Accordingly, relays (RL1 and RL2) will be either on or off. In the above example, relays on time in the first case is 12 hours, 15 minutes and 15 seconds. Similarly, relays off time in the first case is at 12 hours, 16 minutes and 15 seconds. These on/off timings can be changed in the code as per your requirement but with the same format as shown above. You can see each of this on/off time of the relays on the TFT display along with real-time clock (date and time.)

There are two source codes for this project. First code is *DS3231_AC_Scheduler.ino* for Board1 and second is *DS3231_AC_Scheduler_Display.ino* for Board2. First source code is used to set the timings to control the relays at specific time through Board1. The second source code is used to display the time, date, and status of relays on TFT display through Board2. Before compiling and uploading the source codes to respective Arduino boards, you need to include the related libraries such as *RtcDS3231.h*, *TFT_ILI9163C.h* and *Adafruit_GFX.h* in Arduino IDE.

Bill of Materials		Cost
1	DS3231	(\$1)
2	ATMEGA328/UNO	(\$4)
1	128x128 TFT [ILI9163]	(\$3.5)
2	5V sugar-cube relays	(\$1.5)

The total bill of materials cost comes to about \$10 or about ₹ 750, excluding the box, contactor, and 5V power supply.

Aftermath

More circuit ideas on dual sharing of RTC are possible, including in a stroboscope where one microcontroller will create the variable pulse while the other will display the pulse frequency and rotation of shaft, while both the microcontrollers share one RTC. **EFY**

EFY Note
The source code of this project is available for free download at source.efymag.com



Somnath Bera is an avid user of open source software. Professionally, he is a thermal power expert and works as additional general manager